

1. Lab Overview

In this lab, I configure a **DNS client** on Windows 10 and install/configure the **DNS Server role** on Windows Server 2019. The lab focuses on static IP addressing, DNS client configuration, DNS server setup, forward lookup zones, A records, name resolution testing, and DNS traffic analysis using Wireshark.

This lab helps me understand how DNS resolves hostnames to IP addresses and how IT support professionals troubleshoot DNS-related connectivity issues.

2. Business Scenario

This lab simulates a small business or school network where a Windows Server 2019 system provides DNS services for Windows clients.

As the system administrator, I configure the Windows 10 client to use the Windows Server 2019 DNS server. Then I create DNS records on the server and test whether the client can resolve hostnames correctly.

DNS is important in real IT environments because users usually access systems by name instead of IP address. When DNS fails, users may not be able to access websites, servers, file shares, applications, or domain resources.

3. Lab Objectives

The objectives of this lab are to:

- Configure static IP settings on the Windows 10 DNS client.
- Configure static IP settings on the Windows Server 2019 DNS server.
- Install the DNS Server role on Windows Server 2019.
- Open and use DNS Manager.
- Create a forward lookup zone.
- Add host A records.
- Test DNS name resolution from the Windows 10 client.
- Use ping to verify hostname resolution.
- Use Wireshark to capture DNS query and response traffic.
- Understand how DNS maps hostnames to IP addresses.

The original lab goals are to install and configure a DNS server on Windows Server 2019 and configure DNS to resolve hostnames to IP addresses.

4. Lab Environment

Item	Details
DNS Client	Windows 10
DNS Server	Windows Server 2019
Client IP Address	192.168.1.10
Client Subnet Mask	255.255.255.0
Client Default Gateway	192.168.1.1
Client Preferred DNS Server	192.168.1.22
Server IP Address	192.168.1.22
Server Subnet Mask	255.255.255.0
Server Default Gateway	192.168.1.1
Server Preferred DNS Server	127.0.0.1
DNS Zone	jonathan.weissman
Tools Used	DNS Manager, Server Manager, Command Prompt, Wireshark

The Windows 10 client uses the static IP address 192.168.1.10 and preferred DNS server 192.168.1.22. The Windows Server 2019 DNS server uses IP address 192.168.1.22 and preferred DNS server 127.0.0.1.

5. Tools Used

- Windows 10
- Windows Server 2019
- Server Manager
- DNS Manager
- Command Prompt
- Wireshark
- Windows Defender Firewall settings
- IPv4 network adapter settings

6. Steps Performed

Step 1: Configure the Windows 10 DNS Client

I begin by configuring the Windows 10 system with a static IPv4 address.

The Windows 10 client uses:

IP address: 192.168.1.10

Subnet mask: 255.255.255.0

Default gateway: 192.168.1.1

Preferred DNS server: 192.168.1.22

This points the Windows 10 client to the Windows Server 2019 DNS server for name resolution.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: 192 . 168 . 1 . 22

Alternate DNS server:

☐ Validate settings upon exit

Advanced...

OK Cancel

Figure 1: Windows 10 DNS client configured with a static IP address and preferred DNS server.

Step 2: Configure the Windows Server 2019 DNS Server Network Settings

Next, I configure the Windows Server 2019 system with static network settings.

The server uses:

IP address: 192.168.1.22

Subnet mask: 255.255.255.0

Default gateway: 192.168.1.1

Preferred DNS server: 127.0.0.1

The preferred DNS server is set to **127.0.0.1** so the server uses itself for DNS resolution.

The server hostname is also changed to:

WARA-SERVER

Internet Protocol Version 4 (TCP/IPv4) Properties



General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address:

192 . 168 . 1 . 22

Subnet mask:

255 . 255 . 255 . 0

Default gateway:

192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

127 . 0 . 0 . 1

Alternate DNS server:

. . .

☐ Validate settings upon exit

Advanced...

OK

Cancel

Device specifications

Device name	WARA-SERVER
Processor	AMD Ryzen 7 5700U with Radeon Graphics 1.80 GHz (2 processors)
Installed RAM	2.00 GB
Device ID	39584965-0E7D-488F-B1A5-3749AA8BC2AE
Product ID	00454-40000-00001-AA737
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display

Figure 2: Windows Server 2019 configured with a static IP address and local DNS server setting.

Step 3: Install the DNS Server Role

I open **Server Manager** on Windows Server 2019 and use **Add Roles and Features** to install the **DNS Server** role.

The process includes:

1. Opening Server Manager.
2. Selecting **Add Roles and Features**.
3. Choosing the DNS Server role.
4. Adding required features.
5. Confirming installation options.
6. Installing the DNS Server role.

This prepares the server to host DNS zones and records.

The lab instructions show that the DNS Server role is installed using Server Manager and the Add Roles and Features wizard.

Step 4: Open DNS Manager

After installing the DNS Server role, I open **DNS Manager** from Server Manager.

In DNS Manager, I expand the server and review the DNS structure, including:

Forward Lookup Zones

Reverse Lookup Zones

Trust Points

Conditional Forwarders

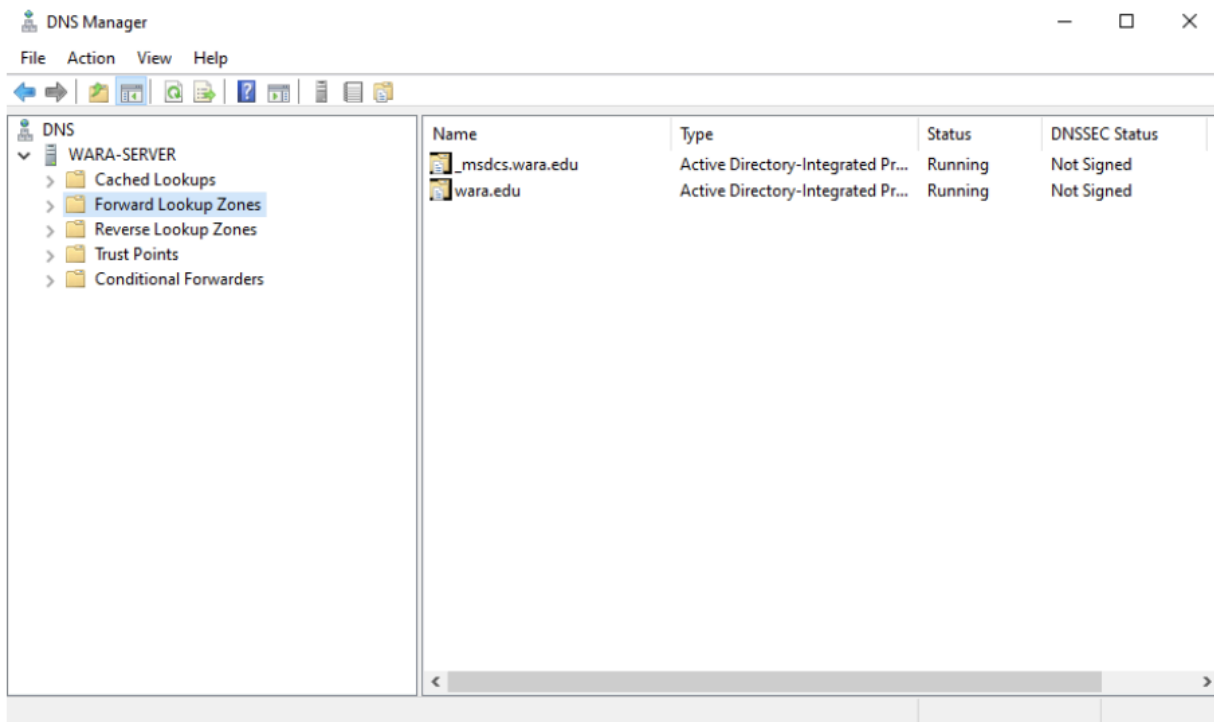


Figure 3: DNS Manager showing the Windows Server DNS environment.

Step 5: Create a Forward Lookup Zone

I create a new forward lookup zone in DNS Manager.

The zone name is:

jonathan.weissman

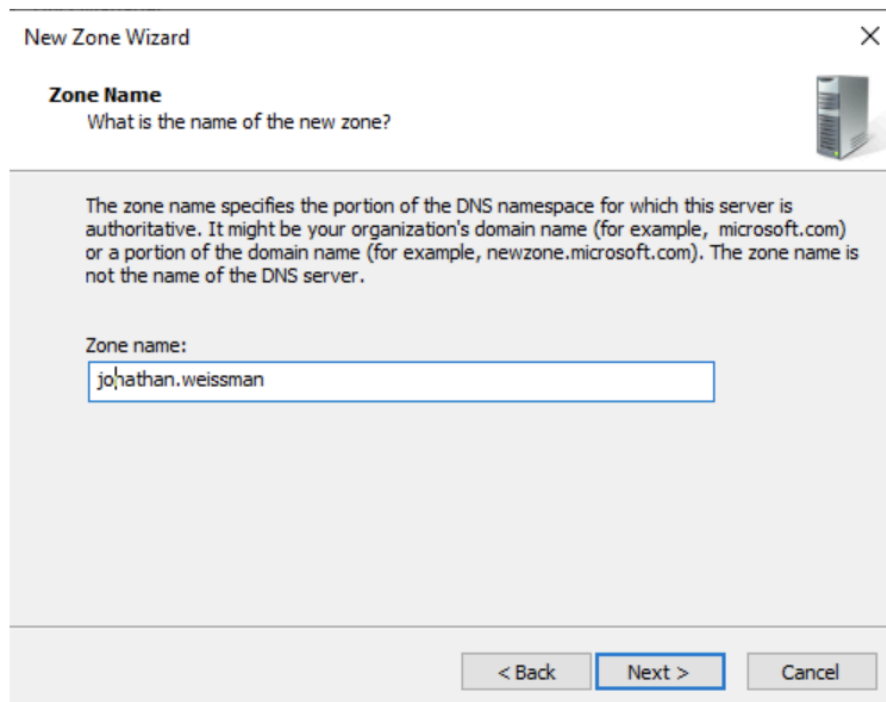
During the New Zone Wizard, I select:

Primary Zone

I also select:

Do not allow dynamic updates

This means DNS records are manually managed instead of automatically updated by clients.



The screenshot shows the 'New Zone Wizard' window at the 'Zone Name' step. The title bar says 'New Zone Wizard' with a close button. Below the title, the section is 'Zone Name' with the question 'What is the name of the new zone?'. To the right is a server icon. The main text explains that the zone name specifies the portion of the DNS namespace for which the server is authoritative, giving examples like 'microsoft.com' or 'newzone.microsoft.com'. Below this, there is a text box labeled 'Zone name:' containing the text 'jonathan.weissman'. At the bottom are three buttons: '< Back', 'Next >', and 'Cancel'.

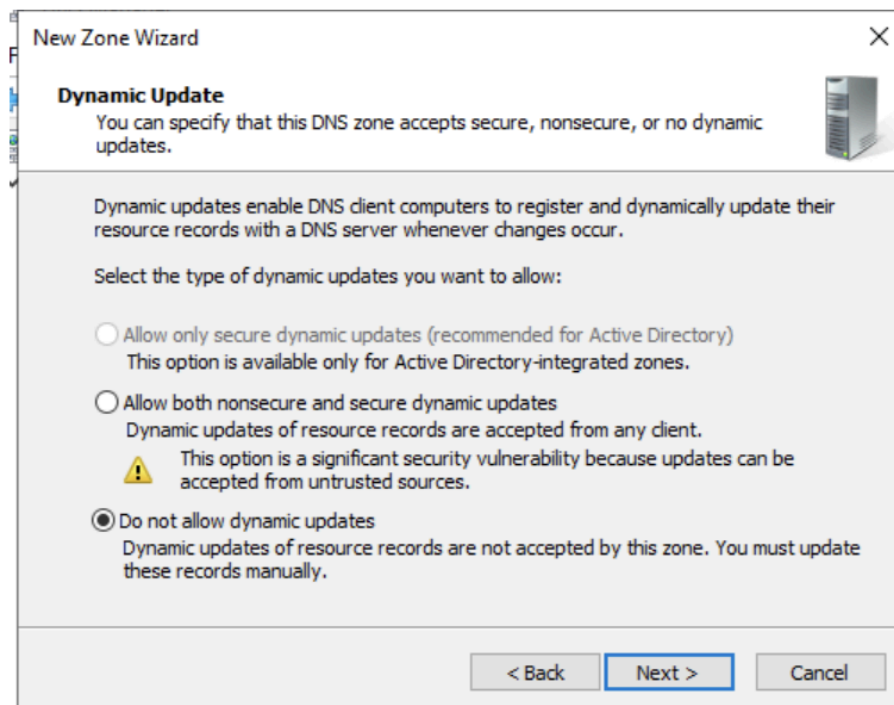
New Zone Wizard

Zone Name
What is the name of the new zone?

The zone name specifies the portion of the DNS namespace for which this server is authoritative. It might be your organization's domain name (for example, microsoft.com) or a portion of the domain name (for example, newzone.microsoft.com). The zone name is not the name of the DNS server.

Zone name:
jonathan.weissman

< Back Next > Cancel



The screenshot shows the 'New Zone Wizard' window at the 'Dynamic Update' step. The title bar says 'New Zone Wizard' with a close button. Below the title, the section is 'Dynamic Update' with the text 'You can specify that this DNS zone accepts secure, nonsecure, or no dynamic updates.' To the right is a server icon. The main text explains that dynamic updates enable DNS client computers to register and dynamically update their resource records. Below this, it asks to 'Select the type of dynamic updates you want to allow:'. There are three radio button options: 'Allow only secure dynamic updates (recommended for Active Directory)' (disabled), 'Allow both nonsecure and secure dynamic updates' (disabled with a warning icon), and 'Do not allow dynamic updates' (selected). At the bottom are three buttons: '< Back', 'Next >', and 'Cancel'.

New Zone Wizard

Dynamic Update
You can specify that this DNS zone accepts secure, nonsecure, or no dynamic updates.

Dynamic updates enable DNS client computers to register and dynamically update their resource records with a DNS server whenever changes occur.

Select the type of dynamic updates you want to allow:

☐ Allow only secure dynamic updates (recommended for Active Directory)
This option is available only for Active Directory-integrated zones.

☐ Allow both nonsecure and secure dynamic updates
Dynamic updates of resource records are accepted from any client.
 This option is a significant security vulnerability because updates can be accepted from untrusted sources.

☒ Do not allow dynamic updates
Dynamic updates of resource records are not accepted by this zone. You must update these records manually.

< Back Next > Cancel

Figure 4: Forward lookup zone created for jonathan.weissman.

Step 6: Add DNS Host A Records

Next, I create host A records inside the forward lookup zone.

The records are:

Hostname	FQDN	IP Address	Purpose
professor	professor.jonathan.weissman	192.168.1.22	Valid server record
rochester	rochester.jonathan.weissman	192.168.1.99	Test record for non-existing host

The **professor** record points to the server, while the **rochester** record points to a non-existing IP address for testing.

Your PDF shows the A records created for **professor** and **rochester**.

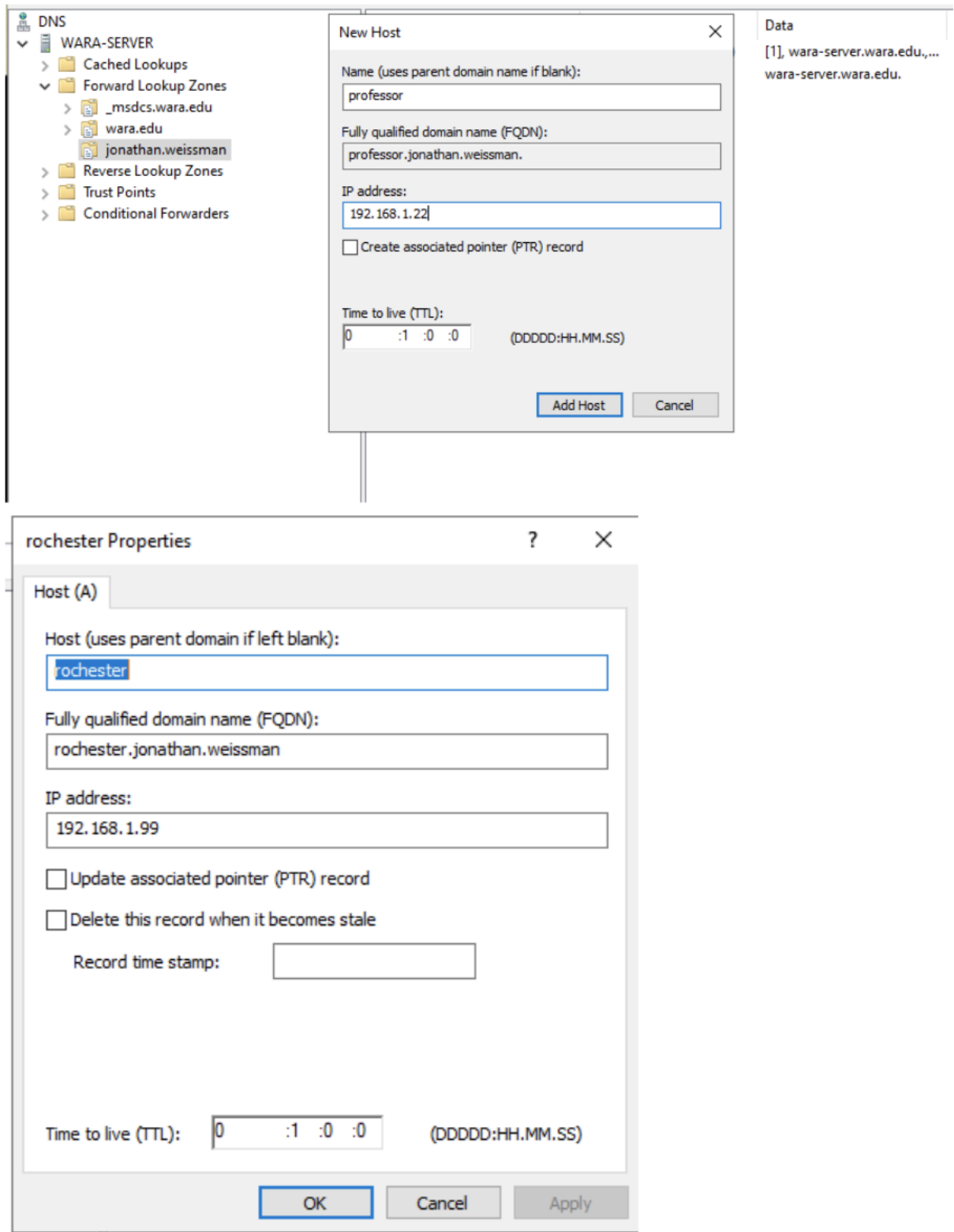


Figure 5: Host A records created for professor and rochester in DNS Manager.

Step 7: Verify DNS Records in DNS Manager

After adding the records, I review the forward lookup zone in DNS Manager. The zone displays the DNS records, including:

professor → 192.168.1.22

rochester → 192.168.1.99

This confirms that the DNS records are created successfully.

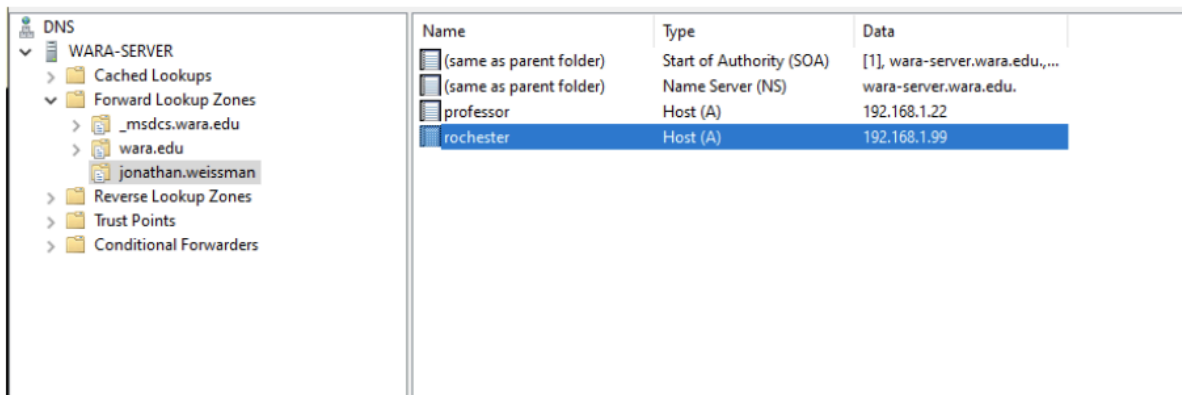


Figure 6: DNS Manager showing created A records inside the forward lookup zone.

Step 8: Test DNS Resolution from the Windows 10 Client

On the Windows 10 client, I open Command Prompt and test name resolution using:

ping professor.jonathan.weissman.

The ping successfully resolves **professor.jonathan.weissman** to:

192.168.1.22

The replies confirm that DNS resolution works for the **professor** record.

Your PDF shows a successful ping to **professor.jonathan.weissman**, resolving to **192.168.1.22**.

```
Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\bbs>ping professor.jonathan.weissman.

Pinging professor.jonathan.weissman [192.168.1.22] with 32 bytes of data:
Reply from 192.168.1.22: bytes=32 time<1ms TTL=128
Reply from 192.168.1.22: bytes=32 time<1ms TTL=128
Reply from 192.168.1.22: bytes=32 time<1ms TTL=128
Reply from 192.168.1.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\bbs>
```

Figure 7: Successful ping test confirming DNS resolution for professor.jonathan.weissman.

Step 9: Capture DNS Query and Response in Wireshark

I use **Wireshark** on the Windows 10 client and apply a DNS filter.

The capture shows DNS query and response traffic for:

professor.jonathan.weissman

This confirms that the client sends a DNS query to the DNS server and receives a DNS response.

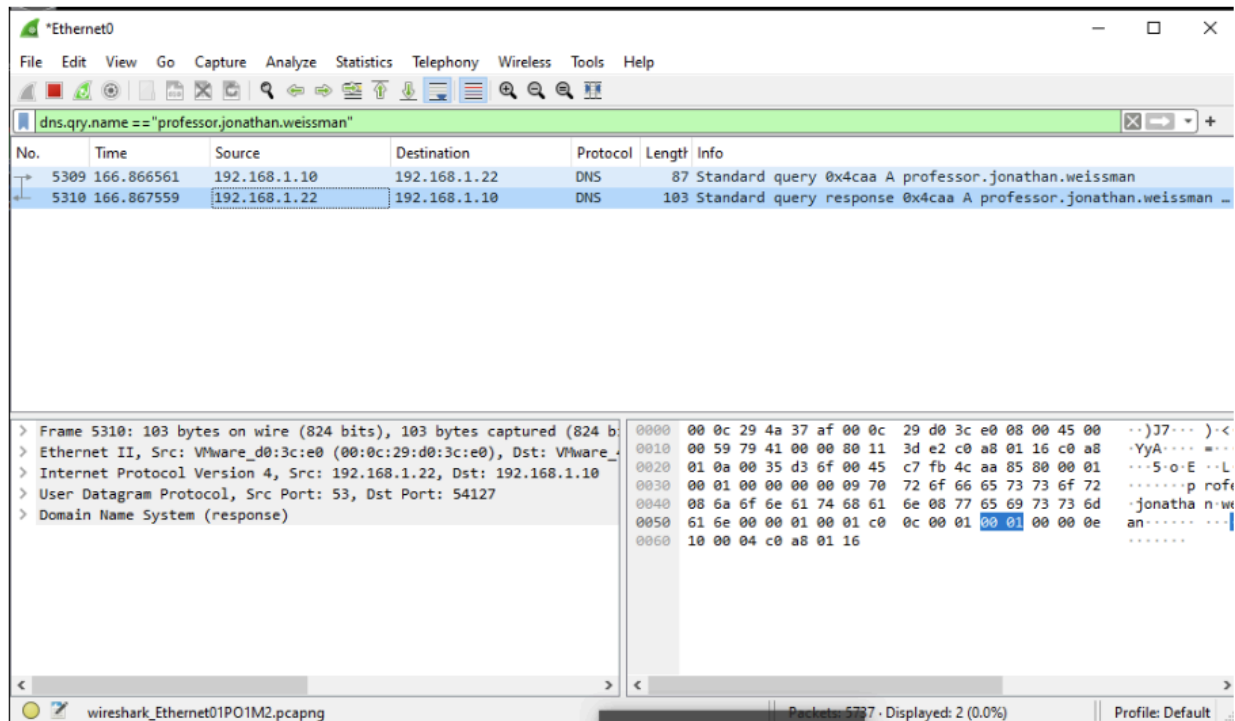


Figure 8: Wireshark captured DNS query and response traffic for professor.jonathan.weissman.

Step 10: Test a DNS Record for a Non-Existing Host

I also test the `rochester` DNS record using:

`ping rochester.jonathan.weissman.`

The hostname resolves to:

`192.168.1.99`

However, the ping fails because the IP address does not belong to an active host. This shows that DNS resolution can work even if the destination host is unreachable.

Your PDF explains that the ping to `rochester.jonathan.weissman` fails, but DNS still maps the name to `192.168.1.99`.

```
Pinging rochester.jonathan.weissman [192.168.1.99] with 32 bytes of data:
Reply from 192.168.1.10: Destination host unreachable.
Reply from 192.168.1.10: Destination host unreachable.
Reply from 192.168.1.10: Destination host unreachable.
Reply from 192.168.1.10: Destination host unreachable.

Ping statistics for 192.168.1.99:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Figure 9: DNS resolves rochester.jonathan.weissman to 192.168.1.99, but the host is unreachable.

7. Key Findings

Finding	Explanation
Static IP addressing is important for DNS servers	DNS clients need a stable DNS server IP address.
DNS clients must point to the correct DNS server	The Windows 10 client uses 192.168.1.22 as its preferred DNS server.
DNS servers can host forward lookup zones	The jonathan.weissman zone is created to manage hostname records.
A records map hostnames to IP addresses	professor resolves to 192.168.1.22, and rochester resolves to 192.168.1.99.
DNS resolution and host reachability are different	rochester resolves correctly, but the ping fails because no active host exists at that IP address.
Wireshark confirms DNS traffic	DNS queries and responses are visible in packet captures.

8. Analysis

This lab shows how DNS allows users and systems to access network resources by hostname instead of IP address. The Windows 10 client points to the Windows Server 2019 DNS server, and the DNS server hosts a forward lookup zone with host records.

The successful ping to professor.jonathan.weissman confirms that DNS resolves the hostname to the correct IP address. The failed ping to rochester.jonathan.weissman is

also useful because it shows an important troubleshooting concept: DNS resolution can succeed even when the destination host is offline or unreachable.

Wireshark provides additional evidence by showing DNS query and response traffic. This connects the command-line test results to actual network traffic.

9. IT Support Relevance

This lab is directly relevant to IT Support Analyst, Help Desk, Desktop Support, Network Support, and Junior System Administrator roles.

It connects to real job tasks such as:

- Configuring static IP addresses
- Setting preferred DNS servers
- Troubleshooting hostname resolution
- Testing DNS with ping
- Understanding DNS zones and records
- Creating and reviewing A records
- Checking whether a DNS issue is actually a connectivity issue
- Using Wireshark to verify DNS traffic
- Supporting Windows client/server network environments

10. Skills Demonstrated

This lab demonstrates:

- Windows 10 network configuration
- Windows Server 2019 DNS configuration
- Static IP addressing
- DNS client setup
- DNS Server role installation
- DNS Manager
- Forward lookup zones
- Host A records
- Name resolution testing
- Command Prompt troubleshooting
- Wireshark DNS traffic analysis
- Client/server troubleshooting
- Technical documentation

11. Ticket-Style Summary

Issue:

A Windows 10 client needs to resolve internal hostnames using a Windows Server 2019 DNS server.

Impact:

Without DNS resolution, users may be unable to access network resources by hostname.

Steps Taken:

- Configure the Windows 10 client with a static IP address.
- Set the Windows 10 preferred DNS server to 192.168.1.22.
- Configure the Windows Server 2019 system with static IP address 192.168.1.22.
- Set the server preferred DNS to 127.0.0.1.
- Install the DNS Server role on Windows Server 2019.
- Open DNS Manager.
- Create a forward lookup zone named jonathan.weissman.
- Disable dynamic updates for the zone.
- Create A records for professor and rochester.
- Test DNS resolution using ping from the Windows 10 client.
- Capture DNS query and response traffic using Wireshark.

Root Cause:

DNS records need to be manually created so the client can resolve internal hostnames.

Resolution:

A DNS forward lookup zone and A records are configured to map hostnames to IP addresses.

Verification:

professor.jonathan.weissman resolves to 192.168.1.22 and responds to ping.
rochester.jonathan.weissman resolves to 192.168.1.99, while the host is unreachable because no active device exists at that IP address.

Status:

Completed successfully.

12. Lessons Learned

In this lab, I learn how DNS clients and DNS servers work together. The client must point to the correct DNS server, and the DNS server must have the correct zone and host records.

I also learn that DNS resolution and network reachability are not the same thing. A hostname can resolve to an IP address even if the target system is offline or unreachable.

This lab helps me understand why DNS is one of the first things IT support teams check when users cannot access websites, servers, file shares, or internal applications.